

Equipment Name	OZONATOR + UV SYSTEM
Protocol Number	AWMSPL-OPM-R0

OPERATION & MAINTENANCE MANUAL

FOR

OZONATION + UV SYSTEM

At

M/s. GINNI FILAMENTS LTD.
ANKLESHWAR

System supplied by

**M/S APURVA WATER MANAGEMENT SYSTEMS
PVT. LTD.**
(A Division of Amey Marketing)

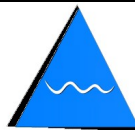
403/A-B-C, SHAHJANAND SHOPPING CENTRE,
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OPERATION MANUAL



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WATER - TODAY'S RESOURCE,
TOMORROW'S TREASURE
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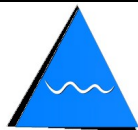
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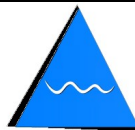
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INTRODUCTION





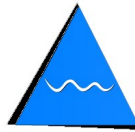
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Nowadays treatment of water occupies the pre-dominant place in the modern industries. Moreover, treatment of water is required for various purposes right from for portable use to industrial applications such as for Pharmaceuticals, food, beverages, lather, textile, and heavy chemical industries. Also, treatment of wastewater especially effluent from toxic disposals attracts attention nowadays due to increase global awareness of ecological & environmental protection.

Treatment of water implicitly means changing its physical & mainly its chemical properties by removing undesirable suspended & dissolved impurities of both organic & inorganic nature. The level of chemical dosing involved & method adopted for treatment are according to the end consumers requirement as well as the nature of composition of raw water sources, which are all mainly from surface wells, lakes, rivers, or from underground sources such as deep wells in some places from sea.

The specific method adopted such as conventional treatment of flocculation /coagulation coagulation, filtration and disinfection, lime soda softening process, permutated filtration, electro dialysis, demineralization, reverse osmosis, and others are depending upon the product quality requirement.

Here, the complete Water Treatment plant is designed for the process application for bulk drug production as well as utility requirement by using R.O. technology; -



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- **PLANT INTRODUCTION:** - Ozonation and UV system is designed on 30 gm/hr for pretreatment system and 40 gm/hr for purified water system.

OZONATION + UV SYSTEM-1 FOR PRE-TREATMENT SYSTEM

01. OZONE RECIRCULATION PUMP-1: - This is required to give the necessary flow and pressure to the system in this case one pump have been provided to do the job.

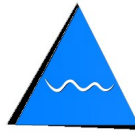
02. OXYGEN GENERATOR WITH OZONE SYSTEM-1: -

Oxygen Generator use two cylinders filled with zeolite as absorbers. As dry oil free compressed air flows through one of the cylinders, the zeolite adsorbs nitrogen. The remaining oxygen passes the cylinder and exits as a product gas. Before the absorber becomes saturated with nitrogen, the feed air is diverted to the second cylinder.

03. UV LAMP FOR RESIDUAL OZONE DISTRUCTION-1: - UV is used for disinfection of bacteria, viruses, molds, algae, and other microorganisms.

04. WATER COOLER-1: - water cooler (if used for cooling only), is a machine that dispenses and often also cools or heats up water with a refrigeration unit.

05. WATER RECIRCULATION PUMP-1: - This is required to give the necessary flow and pressure to the system in this case one pump have been provided to do the job.



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OZONATION + UV SYSTEM-2 FOR PURIFIED WATER SYSTEM

06. OZONE RECIRCULATION PUMP-2: - This is required to give the necessary flow and pressure to the system in this case one pump have been provided to do the job.

07. OXYGEN GENERATOR WITH OZONE SYSTEM-2: -

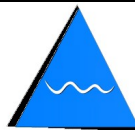
Oxygen Generator use two cylinders filled with zeolite as absorbers. As dry oil free compressed air flows through one of the cylinders, the zeolite adsorbs nitrogen. The remaining oxygen passes the cylinder and exits as a product gas. Before the absorber becomes saturated with nitrogen, the feed air is diverted to the second cylinder.

08. UV LAMP FOR RESIDUAL OZONE DISTRUCTION-2: - UV is used for disinfection of bacteria, viruses, molds, algae, and other microorganisms.

09. WATER COOLER-2: - water cooler (if used for cooling only), is a machine that dispenses and often also cools or heats up water with a refrigeration unit.

10. WATER RECIRCULATION PUMP-2: - This is required to give the necessary flow and pressure to the system in this case one pump have been provided to do the job.

OPERATION MANUAL



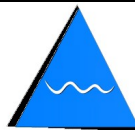
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TECHNICAL SPECIFICATION MECHANICAL





Equipment Name OZONATOR + UV SYSTEM

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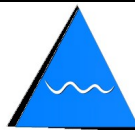
OZONATION + UV SYSTEM-1 FOR PRE-TREATMENT SYSTEM**• OZONE RECIRCULATION PUMP-1**

Quantity	1 NO.
Capacity	2.0-2.5 m3/hr.
Discharge pressure	2.0 Kg/cm ²
Type	Centrifugal
MOC	SS 316
Model	HYGINOX 17 SE 28
MAKE	Inoxpa

• OZONATOR-1

Quantity	One No.
Working Capacity	30 Gram / hr.
Body MOC	SS 304
Type	Corona Discharge
Cell MOC	SS 316 - Chilled Water Cooled
No. Of Cells	2 Nos. 2 ft. long Water cooled
Power Supply	High Frequency High Voltage Circuit Independent to each cell.
Instruments	Ozone Indicator, Rota Meter.
Indicators Lamps	Power, Fuse, Ozone On
Mounting	SS Trolley Mounted
Diffuser	SS 316 Venturi Ejector, 1" x 6" long
Weight	Approximate 70 Kg.
Size (Approx)	20" W X 20" D X 36 "H
Ozone Out Put	30 Gram / hr.
Oxygen Flow	8 Lpm
All Contact Parts	In SS 316
Ozone Tube	Silicon /Teflon 1/4"

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Equipment Name OZONATOR + UV SYSTEM

Protocol Number AWMSPL-OPM-R0

	5 meter long
Supply Fuse	5 Amp. / MCB 20 Amp.
Power	600 W
Make	Aqua zone / Equiv.

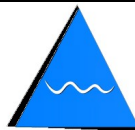
• OXYGEN GENERATOR-1

Quantity	1 NO.
MOC	SS 304.
Type	Industrial
Power	550 Watt.
Weight	30 Kgs
Oxygen Flow	8 LPM
Size	24" * 15" * 12" Approx.
Oxygen Purity	92 %
Oxygen Pressure	14 PSI
Make	Aqua zone / Equiv.

• WATER RECIRCULATION PUMP-1

Quantity	1 NO.
Capacity	2.0 m3/hr.
Discharge pressure	2.0 Kg/cm ²
Type	Centrifugal
MOC	CI/ SS 304
Model	SMSJ-103
MAKE	LUBI





Equipment Name OZONATOR + UV SYSTEM

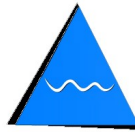
Protocol Number AWMSP-OPM-R0

OZONATION + UV SYSTEM-2 FOR PURIFIED WATER SYSTEM**• OZONE RECIRCULATION PUMP-2**

Quantity	1 NO.
Capacity	2.0-2.5 m3/hr.
Discharge pressure	2.0 Kg/cm ²
Type	Centrifugal
MOC	SS 316
Model	HYGINOX 17 SE 28
MAKE	Inoxpa

• OZONATOR-2

Quantity	One No.
Working Capacity	40 Gram / hr.
Body MOC	SS 304
Type	Corona Discharge
Cell MOC	SS 316 - Chilled Water Cooled
No. Of Cells	2 Nos. 2 ft. long Water cooled
Power Supply	High Frequency High Voltage Circuit Independent to each cell.
Instruments	Ozone Indicator, Rota Meter.
Indicators Lamps	Power, Fuse, Ozone On
Mounting	SS Trolley Mounted
Diffuser	SS 316 Venturi Ejector, 1" x 6" long
Weight	Approximate 70 Kg.
Size (Approx)	20" W X 20" D X 36 "H
Ozone Out Put	40 Gram / hr.
Oxygen Flow	10 Lpm
All Contact Parts	In SS 316
Ozone Tube	Silicon /Teflon ¼" 5 meter



Equipment Name	OZONATOR + UV SYSTEM
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	long
Supply Fuse	5 Amp./ MCB 20 Amp.
Power	750 W
Make	Aqua zone / Equiv.

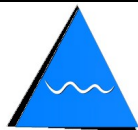
• OXYGEN GENERATOR-2

Quantity	1 NO.
MOC	SS 304.
Type	Industrial
Power	750 Watt.
Weight	30 Kgs
Oxygen Flow	8-10 Lpm
Size	24" * 15" * 12" Approx.
Oxygen Purity	92 %
Oxygen Pressure	14 PSI
Make	NUVO

• WATER RECIRCULATION PUMP-2

Quantity	1 NO.
Capacity	2.0 m3/hr.
Discharge pressure	2.0 Kg/cm ²
Type	Centrifugal
MOC	CI/ SS 304
Model	SMSJ-103
MAKE	LUBI

OPERATION MANUAL



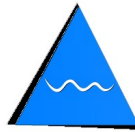
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VALVES





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- BALL VALVE**

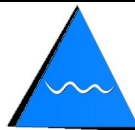
Quantity	06 Nos.
MOC	PVC
End Connection	SOLUTION END
Operating pressure	6 kg/cm ²
Operating temperature	Ambient
Size	1.0" (25 NB)
Location	Pretreatment Ozonator system
Make	AWMS

- BALL VALVE**

Quantity	02 Nos.
MOC	PVC
End Connection	SOLUTION END
Operating pressure	6 kg/cm ²
Operating temperature	Ambient
Size	1.0" (25 NB)
Location	Purified water Ozonator system
Make	AWMS

- SAMPLE VALVE**

Quantity	3 No.
MOC	SS 316
End Connection	TC end
Operating pressure	6 kg/cm ²
Operating temperature	Ambient
Size	1.0"
Location	Pretreatment Ozonator system + Purified water



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	system
Make	AWMS

• SAMPLE VALVE

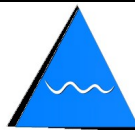
Quantity	2 Nos.
MOC	UPVC
End Connection	Threaded end
Operating pressure	6 kg/cm ²
Operating temperature	Ambient
Size	8 MM
Location	Purified water Ozonator system
Make	AWMS

• DIAPHRAGM VALVE

Quantity	3 No.
MOC	SS 316
End Connection	TC end
Operating pressure	6 kg/cm ²
TYPE	EPDM
Size	1.0"
Location	Purified water Ozonator system

• DIAPHRAGM VALVE

Quantity	1 No.
MOC	SS 316
End Connection	TC end
Operating pressure	6 kg/cm ²
TYPE	EPDM
Size	1.5"



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Location	Purified water Ozonator system
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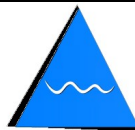
INSTRUMENTS

- PRESSURE GUAGES**

Quantity	2 Nos.
Case	SS 316L
End Connection	THREADED
Entry	Bottom entry
Range	0-7 kg/cm ²
Lenses	Glass
Pressure element	Bourdon
Accuracy	± 1% of F.S.

- PRESSURE GUAGES**

Quantity	1 No.
Case	SS 316L
End Connection	TC END
Entry	Bottom entry
Range	0-7 kg/cm ²
Lenses	Glass
Pressure element	Bourdon
Accuracy	± 1% of F.S.



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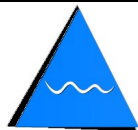
• ORP ANALYZER

Quantity	04 Nos.
Fluid state	Liquid
Temperature range	50° C
Type of Analysis	ORP
Model	605-C
ORP Range	-1999 mV + 1999 mV
Output	RELAY OUTPUT
Sensor connection	Threaded end
Accuracy	1% of F. S
MAKE	DIGITAL

• ELECTRIC PANEL-1

Quantity	02 Nos.
Type	Skid mounted
MOC	MS POWDER COATED
Input voltage	415 VAC
Location	RO Skid
Input HZ	50 HZ
Size	700 H x 600 W x 250 D
Make	AWMS

OPERATION MANUAL



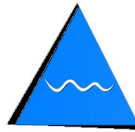
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OPERATION

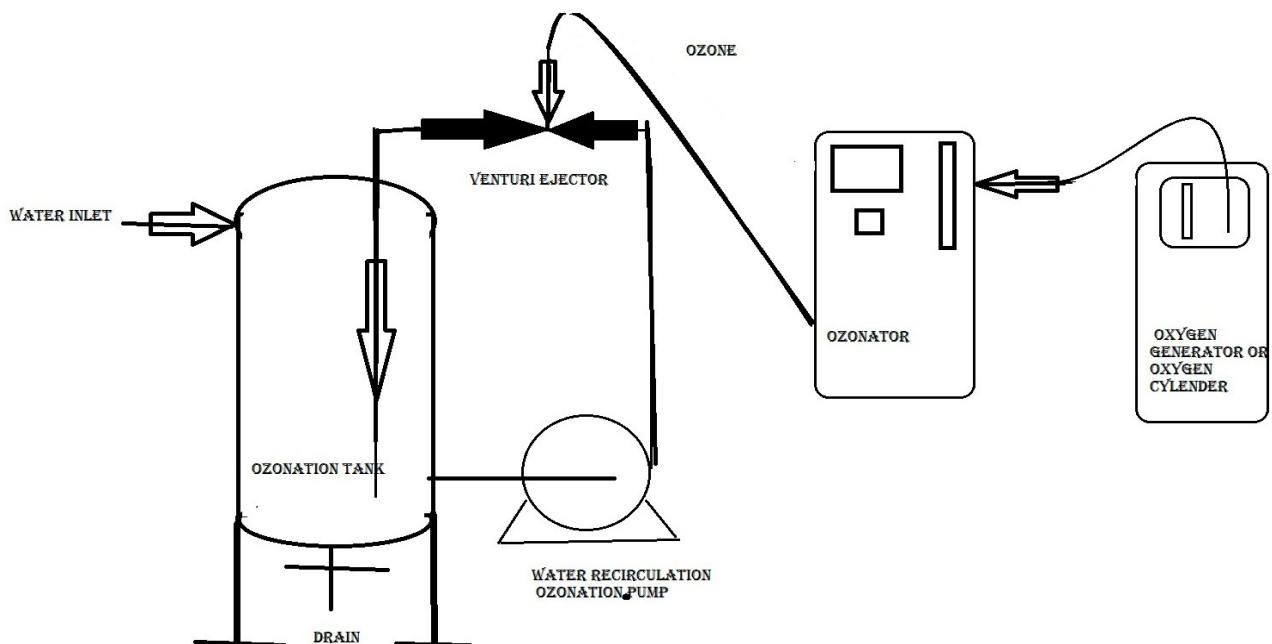


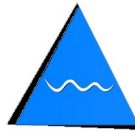


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01. OZONE RECIRCULATION PUMP: - Start feed Water Supply Pump for feeding water to Filtered water storage tank. Open delivery valve gradually and ensure the minimum pressure of 1.5-3.0 Kg/cm².

02. OZONATOR WITH OXYGEN GENERATOR: - Oxygen Generator use two cylinders filled with zeolite as absorbers. As dry oil free compressed air flows through one of the cylinders, the zeolite adsorbs nitrogen. The remaining oxygen passes the cylinder and exits as a product gas. Before the absorber becomes saturated with nitrogen, the feed air is diverted to the second cylinder.





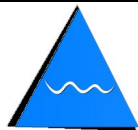
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- **START UP OPERATION:** - Please refer to Front over of Ozonator.
- **SHUT - OFF OPERATION:** - Please refer to Front Over of Ozonator
- **GENRAL MAINTENANCE:** - Please refer to Front Over of Ozonator
- **PRECAUTIONS:** - Please refer to Front Over of Ozonator

This process is then repeated in the second vessel to complete a cycle that allows to oxygen generator to deliver a constant flow of product oxygen at 90 % minimum purity. Under Normal operating conditions, the molecular sieve is completely regenerative and will last indefinitely.

03. WATER RECIRCULATION PUMP: - Start feed Water Supply Pump for feeding water to Oxygen generator. Open delivery valve gradually and ensure the minimum pressure of 1.5-3.0 Kg/cm².

OPERATION MANUAL



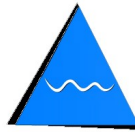
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MAINTENANCE & TROUBLE SHOOTING





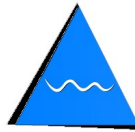
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SYSTEM MAINTENANCE ...

This section consists of summary of system maintenance tasks which include schedule maintenance, trouble shooting and repairs. This section is to be read in conjunction with the detailed maintenance/trouble shooting/ repair instructions in the respective manuals supplied by various manufacturers. All maintenance should be documented in a maintenance log sheet, as to maintain a chronological history of all maintenance tasks performed. While attempt has been made to make this section as exhaustive as possible, site-specific problems are not included here. Need to evolve and establish detailed maintenance schedule specific to the site may exist and instruction for handling such situations may be added to this section, in consultation with manufacturer, where necessary.

ROUTINE MAINTENANCE:

Open the valve bodies periodically and the condition of internal parts and lubricate the seats/spindles.



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RECOMMENDED TEST KITS FOR REGULAR WATER ANALYSIS:

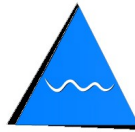
- PH Scan.
- Ozone Test Kit.

TO DO: -

- To follow the guidelines provided in OPM Manual for proper starting and shut down of the plant.
- The plant to be operated by the trained and experienced personnel in Ozonator system.
- Unit should be kept two feet above water level.
- Unit should be kept in cool & dry place away from moisture.
- Serviceable parts return to authorized service center for servicing.
- Do not use without external earthing.

DO NOT: -

- Do not immerse in water.
- Avoid ozone exposure to human being.
- No user serviceable parts inside.
- Warranty void if seal is found broken.



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Can Do for You.....

EFFICIENT FIELD SERVICE

We maintain a highly trained staff of engineers to service your equipments for maintenance, or to assist you, should emergencies ever occur.

COMPLETE REPAIR SERVICE

Our engineers will repair your equipment as per Specifications, using only genuine spare parts.

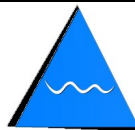
SPARE PARTS

By stocking genuine Apurva spare parts, we can help you avoid costly delays, or substituting inferior parts. Using genuine spare parts on your Apurva equipment will help to keep even older machinery running in good as new condition Apurva's prompt factory shipment to ensure you on time delivery.

AMC AND O&M CONTRACTS

We undertake the Annual Maintenance and Operation & Maintenance contracts for trouble free continuous plant operation and maintenance by using our technically qualified technicians and engineers in the same field.

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OPERATION MANUAL

M/s. GINNI FILAMENTS LTD.

OZONATOR + UV SYSTEM

REPORT APPROVAL FOR CLOSE – OUT

**M/S. GINNI FILAMENTS
LTD.**

DESIGNATION

SIGNATURE

DATE

Reviewed By:

Reviewed By:

Approved By:

Approved By:

